

Theorem. $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$

Levels recommended for proof: 5

Proof. Recall De Moivre's theorem, which says that $(\cos(\theta) + i\sin(\theta))^n = \cos(n\theta) + i\sin(n\theta)$. Now we have that

$$\begin{aligned} \frac{(\cot(\theta) + i)^{2n+1} - (\cot(\theta) - i)^{2n+1}}{2i} &= \frac{(\cos(\theta) + i\sin(\theta))^{2n+1} - (\cos(\theta) - i\sin(\theta))^{2n+1}}{2i(\sin^{2n+1}(\theta))} \\ &= \frac{\cos((2n+1)\theta) + i\sin((2n+1)\theta) - \cos((2n+1)\theta) + i\sin((2n+1)\theta)}{2i(\sin^{2n+1}(\theta))} = \frac{\sin((2n+1)\theta)}{\sin^{2n+1}(\theta)}. \end{aligned}$$

Now let's suppose that $\theta = \frac{m\pi}{2n+1}$ for some integer m that is between 1 and n inclusive. Then we know that $\sin((2n+1)\theta) = 0$ and therefore $\frac{(\cot(\theta) + i)^{2n+1} - (\cot(\theta) - i)^{2n+1}}{2i} = 0$. Now I will use the binomial theorem to expand $(\cot(\theta) + i)^{2n+1} - (\cot(\theta) - i)^{2n+1}$, which we know must be equal to 0. We will get

$$0 = \sum_{r=1}^{2n+1} \binom{2n+1}{r} \cot^r(\theta) i^{2n+1-r} - \sum_{r=0}^{2n+1} \binom{2n+1}{r} \cot^r(\theta) (-i)^{2n+1-r} = \sum_{r=1}^{2n+1} \binom{2n+1}{r} \cot^r(\theta) [i^{2n+1-r} - (-i)^{2n+1-r}].$$

When r is odd, $2n+1-r$ is even, so the bracketed term will look like $i^{2k} - (-i)^{2k}$ where k is an integer, but this equals $(-1)^k - (-1)^k = 0$. Therefore the only terms that survive are the terms where r is even, so we can replace r with s ranging from 1 to n where s is $2r$. We have

$$\sum_{s=0}^n \binom{2n+1}{2s} \cot^{2s}(\theta) [i^{2(n-s)+1} - (-i)^{2(n-s)+1}]$$

Let's replace s with $n-s$, and note the identity $\binom{2n+1}{2s} = \binom{2n+1}{2n+1-2s}$ (since we can choose $2s$ things in the same number of ways we can exclude $2s$ things). Let's call $n-s$ k , then we have

$$\sum_{k=0}^n \binom{2n+1}{2k+1} \cot^{2(n-k)}(\theta) [i^{2k+1} - (-i)^{2k+1}]$$

This last term is $i + i = 2i$ when k is even and $-i - i = -2i$ when k is odd. Since the whole thing equals 0, we can divide it through by $2i$ to get the following:

$$\sum_{k=0}^n \binom{2n+1}{2k+1} \cot^{2(n-k)}(\theta) (-1)^k = 0$$

If $x = \cot^2(\theta)$ we can write the sum above as

$$\binom{2n+1}{1} x^n - \binom{2n+1}{3} x^{n-1} + \dots + (-1)^n = 0$$

Note that the roots of the polynomial above are exactly the n numbers $\cot^2\left(\frac{m\pi}{2n+1}\right)$ for m an integer going from 1 to n . The same logic applies if m is an integer greater than n or less than 0, however this would give one of the same numbers – this can be argued from symmetry of the graph, or from the fact that it must be the case as $\cot^2\left(\frac{m\pi}{2n+1}\right)$ for m from 1 to n is always distinct as $\cot$ is increasing in that range, and an n -degree polynomial cannot have more than n roots (or else you could factor it more than n times by the factor theorem).

Now we know that the sum of roots of a polynomial is minus the second leading coefficient divided by the leading coefficient. Applying this to the polynomial above gives

$$\sum_{m=1}^n \cot^2\left(\frac{m\pi}{2n+1}\right) = \frac{\binom{2n+1}{3}}{\binom{2n+1}{1}} = \frac{\frac{(2n+1)!}{3!(2n-2)!}}{2n+1} = \frac{(2n)!}{6(2n-2)!} = \frac{(2n)(2n-1)}{6} = \frac{n(2n-1)}{3}$$

We now need two facts:

1. For θ between 0 and $\frac{\pi}{2}$, $\sin(\theta) = \int_0^\theta \cos(x)dx$. Since $\cos(x)$ is always between 0 and 1 in that range, it means that $\sin(\theta)$ is between 0 and θ
2. For θ between 0 and $\frac{\pi}{2}$, $\tan(\theta) = \int_0^\theta \sec^2(x)dx$ (you can write $\tan$ as $\frac{\sin}{\cos}$ and differentiate it using the quotient rule and verify that it is $\sec^2$). Since $\cos(x)$ is always between 0 and 1 in that range, it means that $\sec(\theta)$ is greater than 1 for all non-zero values in that range, and therefore $\tan(\theta) > \theta$

So $0 < \sin(\theta) < \theta < \tan(\theta)$ for θ between 0 and $\frac{\pi}{2}$. Since in $\sin(\theta) < \theta < \tan(\theta)$ they are all positive, taking the reciprocal of both sides flips the inequality, so $\cot(\theta) < \frac{1}{\theta} < \csc(\theta)$. We still have positive things, so we will square both sides to get $\cot^2(\theta) < \frac{1}{\theta^2} < \csc^2(\theta)$. However we can say something else, which is that $\cot^2(\theta) < \frac{1}{\theta^2} < 1 + \cot^2(\theta)$, as $\csc^2(\theta) = \frac{1}{\sin^2(\theta)} = \frac{\sin^2(\theta) + \cos^2(\theta)}{\sin^2(\theta)} = 1 + \cot^2(\theta)$. Now

$$\sum_{m=1}^n \cot^2\left(\frac{m\pi}{2n+1}\right) < \sum_{m=1}^n \frac{(2n+1)^2}{m^2\pi^2} < \sum_{m=1}^n 1 + \cot^2\left(\frac{m\pi}{2n+1}\right) = n + \sum_{m=1}^n \cot^2\left(\frac{m\pi}{2n+1}\right)$$

But we know

$$\sum_{m=1}^n \cot^2\left(\frac{m\pi}{2n+1}\right) = \frac{n(2n-1)}{3}$$

Therefore

$$n + \sum_{m=1}^n \cot^2\left(\frac{m\pi}{2n+1}\right) = \frac{n(2n+2)}{3}$$

Therefore we have the tight bound

$$\frac{n(2n-1)}{3} < \sum_{m=1}^n \frac{(2n+1)^2}{m^2\pi^2} < \frac{n(2n+2)}{3}$$

for all positive integers n. Lets multiply through by $\frac{\pi^2}{(2n+1)^2}$ to get that

$$\frac{n(2n-1)\pi^2}{3(2n+1)^2} < \sum_{m=1}^n \frac{1}{m^2} < \frac{n(2n+2)\pi^2}{3(2n+1)^2}$$

But notice that as n gets large, $\frac{n(2n-1)}{3(2n+1)^2}$ and $\frac{n(2n+2)}{3(2n+1)^2}$ will both approach $\frac{1}{6}$, as they are exactly $\frac{1}{6}$ save for the +1 and +2 and -1 terms which contribute a factor on the order of $\frac{1}{n}$. Therefore the sum $\sum_{m=1}^n \frac{1}{m^2}$ is bounded below and above by something that approaches $\frac{\pi^2}{6}$, and therefore the infinite sum is equal to exactly $\frac{\pi^2}{6}$.

□